

HUGEN 2073 case study 2 exercises

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Introduction

You will work in small groups to (approximately) replicate some post-GWAS analyses from the anxiety GWAS paper you recently read, and then briefly share your results (or lack thereof) with the class.

Here is the paper:

Strom, Nora I et al. "Genome-wide association study of major anxiety disorders in 122,341 European-ancestry cases identifies 58 loci and highlights GABAergic signaling." medRxiv : the preprint server for health sciences 2024.07.03.24309466. 5 Jul. 2024, doi:10.1101/2024.07.03.24309466. Preprint.

Note that you will probably need to consult both the paper and the supplementary information. You might also find useful some of the other supporting information linked to in the paper.

You are encouraged to use AI to help you. I would recommend uploading the article pdf into your LLM of choice and asking it to guide you throughout. Note that in some places the authors have made their code available. You may find helpful this document containing some of their

Setup

Log on to <http://hugen.crc.pitt.edu>, open a terminal window, and start an interactive job with `srun -M teach -A hugen2073-2026s --pty --time=01:15:00 bash`.

The GWAS summary stats have been downloaded for you in `/ix1/hugen2073-2026s/case_study`.

- The main results file is `ANX_2026_daner_fullANX_v12_woUTAH_11022026.gz`
- You should also check the README file `readme_ANX_2026_GWAS_sumstats.txt`

Analyses

After either choosing or being assigned to one of the analyses below, see how well you can replicate some part of it.

PheWAS

In the “Genetic overlap between ANX and other phenotypes” section, the authors performed PheWAS for a subset of their SNPs using an R package. See if you can use the same R package to reproduce some of the results they found for those SNPs. Note you will probably need to install the package.

GSMR

In the “Genetic overlap between ANX and other phenotypes” section, the authors also performed bi-directional generalized SMR (GSMR) with another R package. (You may need to try downloading summary stats from *other* GWAS to do this.)